

Pension Systems at the Edge: Governance a Real Actuarial Risk

The 2025 Autumn Governance Series: Pension Fund Governance Focus



TANYA MATANDA

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Across advanced economies, pension systems are entering a structural

test. Gender gaps, demographic collapse, climate volatility, and wealth concentration are no longer social debates—they are measurable distortions in actuarial balance sheets. Without corrective governance, solvency tests will extend by up to nine years, dependency ratios will halve, and wealth will metastasize into fiscal liabilities.

This is not about fairness. It's about arithmetic. The stability of pension systems—whether public or occupational—depends on whether board actuaries, and regulators treat these structural inequities as quantifiable rather than political abstractions.

The global data tell a consistent story:

- **Gender:** Women retire with roughly **26 % less pension income** across OECD countries, cutting contribution density and heightening long-term risk.
- **Demography:** Worker-to-retiree ratios are collapsing from historic highs of 16 : 1 to as low as 1.4 : 1 by mid-century.
- **Climate:** Funding horizons lengthen **three to nine years** in moderate warming scenarios, stretching liabilities while compressing asset returns.
- **Wealth and Health:** Mortality differentials of ten to twenty years across income strata undermine the basic actuarial assumption of uniform longevity improvement.

Governance, not generosity, is the hinge. The next decade will determine whether pension systems evolve into capital-efficient institutions or descend into structural insolvency disguised as policy compassion.

The Arithmetic of Imbalance

Actuarial neutrality once meant matching assets to liabilities. That neutrality now hides systemic error. Models still assume continuous employment, uniform mortality, and stable participation—conditions that no longer

In OECD economies, women perform 75 % of unpaid care and dominant time work. Their careers average seven years shorter than men's, producing a 26 % lifetime contribution gap and a 37 % smaller private pension potential. cumulative drag: nearly **\$3 trillion** in unpriced liabilities by 2030.

Longevity, too, is asymmetric. Germany's Heubeck 2018 G tables make life expectancy a function of income decile—a recognition that low-income

live roughly six years fewer than affluent peers. Failure to model this difference misallocates reserves and distorts contribution policy.

In the United States, Hispanic households hold only 23 % of white household retirement wealth, Black households 43 %. Participation gaps propagate through contribution flows and investment behavior, yet most plans still model “average” entrants. Governance failure here is informational: systems are managing to outdated averages in a world of demographic fragmentation.

Climate Change as Actuarial Drag

Climate volatility now functions as a slow-motion solvency shock. The Institute and Faculty of Actuaries' *Planetary Solvency Framework* projects 10 % GDP contraction by late century under high-emission paths. Defined benefit schemes would see funding ratios fall 40 %; defined-contribution accounts lose 30 %+ of value.

The Canadian Actuaries Climate Index recorded ten straight quarters of record highs by 2020. Germany's DAV Climate Day 5.0 linked each degree of temperature increase to a 2–4 % rise in heat-related mortality among the elderly. Extended morbidity shortens working lives and lengthens payout periods.

Boards treating climate disclosure as optics rather than liability management are mis-pricing time itself. Funding deficits widen not because market but because temperature and mortality curves diverge.

Demography: The Inevitable Compression

Pay-as-you-go systems once thrived on demographic leverage—many workers supporting few retirees. That leverage has inverted.

The IMF's *Proportionality Measure* (PM) defines actuarial fairness where lifetime benefits equal lifetime contributions (PM = 1). Pre-2007 Europe operated near $PM \approx 2.0$ —unsustainable. Reforms reduced that by 25 % for younger cohorts but largely insulated current retirees.

Canada's 1997 CPP reform remains the model: partial funding, 70 % from contributions and 30 % from investment income, keeping long-term stable at a 9.9 % rate. Absent similar automatic balancing mechanisms, Sweden's notional defined-contribution system—most countries will exhaust political room for adjustment before the math adjusts itself.

Disability and Health: The Silent Liability

Disability incidence introduces nonlinear costs. The U.S. Individual Disability Income dataset (1990–2007) shows termination rates varying **up to 212 %** between diagnostic groups. Each 1 % rise in unemployment raises female disability claims 8.5 %.

In South Africa, the 2022 Life and Disability Shortfall Study revealed a **trillion** insurance gap and mortality improvements stalling for low-income cohorts. Many African life tables still rely on 1980s data, producing structural mis-pricing of risk.

Canada's 2019 CPP integration of post-retirement disability benefits is counter-example: one valuation framework pricing health shocks into contribution logic. Where governance separates health data from financial models, solvency estimates drift into fiction.

Equity as Efficiency

Equity isn't sentiment; it's solvency hygiene.

Germany's caregiver credits (three pension years per child) narrowed the gender gap by seven percentage points and cut old-age poverty among women by 40 %. Sweden's *Barnårsrätt* credits cost about 2 % of expenses yet stabilized female participation—a direct boost to contribution density. France's €24 billion caregiver credit system offsets far larger future welfare costs.

In actuarial terms, inclusion is capital efficiency: broader participation, longer contribution periods, lower fiscal backstops. Systems that model equity perform better because they measure reality, not ideals.

Governance Architecture: From Board Ritual to Risk Engine

The temporal mismatch defines modern pension failure: boards think in quarters; liabilities compound over decades. Bridging that gap means transforming governance from ritual into risk discipline—by asking the right questions.

a. Risk Committees with Actuarial Independence

Do we have an independent mechanism to interrogate valuation assumptions?

or are we relying on the same internal models that produce them? How are those assumptions externally benchmarked? Independence is not decorative; it's protection against optimism bias and political pressure. Without it, boards govern forecasts, not realities.

b. Equity-Adjusted Reporting

Are we managing to averages that conceal actuarial risk? Do we disaggregate funding and participation by gender, race, and income? How many basis points of solvency uncertainty stem from unmodeled inequity? Data blindness is balance-sheet risk.

c. Climate Integration

Have we embedded TCFD-aligned scenarios into solvency models, or confined them to ESG appendices? What does a one-degree temperature increase mean for our funding duration and mortality tables? Is “climate beta” visible on our risk dashboard—or missing by habit?

d. Intergenerational Stress Testing

When we simulate proportionality across cohorts, are we transferring liabilities forward while declaring solvency today? A system sustainable for one generation but punitive for the next may meet legal standards yet lack actuarial consistency. What self-correcting triggers exist to rebalance in crisis?

e. Incentive Realignment

Are trustee and executive incentives tied to 30-year funding health or short-term performance? If compensation rewards short-term returns while liabilities accrue over decades, the system is misaligned by design. Performance metrics that ignore long-term sustainability risks create a dangerous disconnect between what is rewarded and what is required.

solvency must become a generational KPI.

Boards able to answer these questions credibly demonstrate fiduciary maturity—not compliance theater but control.

Market Signal: Governance as Cost of Capital

Markets already price governance quality. Credit agencies fold pension underfunding into sovereign spreads; a single-notch downgrade for pension exposure can add **40 basis points** to borrowing costs.

Funds with transparent governance and integrated climate metrics—Norway's GPFG, the Netherlands' ABP, Canada's CPPIB—consistently outperform. Their advantage isn't ideology but information: consolidated boards, independent actuarial audits, and clear accountability chains.

Governance maturity correlates with return stability; opacity correlates with volatility. In financial terms, fairness reduces beta.

Regional Contrasts

United States – Fragmented systems and \$1.4 trillion in unfunded liabilities make standardization urgent. The SEC's forthcoming climate-risk rules will serve as the first step toward unified solvency disclosure.

Europe – Northern systems emphasize automatic stabilizers; Southern systems face demographic inertia. The EU's IORP II and SFDR frameworks are forcing pension governance to merge risk oversight with sustainability metrics, creating an emerging benchmark for fiduciary convergence.

Asia – Aging without full coverage. Japan's GPIF integrates ESG data in actuarial baseline, effectively redefining fiduciary duty as systemic stal

Africa & Latin America – The limiting factor is data. Life tables and dis statistics are decades old; without modern demographic data, risk pric solvency management remain conjectural.

The Political Economy of Pension Risk

Reform inertia thrives on invisibility. Pension liabilities grow silently un appear in fiscal crises—Greece, Illinois, Brazil. By the time voters notic compounding has outpaced policy capacity.

Sustainable systems embed automaticity: Sweden's NDC and Canada's funding model depoliticize adjustment by codifying proportionality. Th of delay compounds faster than interest; governance is the only credit emptive instrument.

The Next Decade: Metrics That Matter

Credible governance in the 2026–2036 cycle will hinge on three quant indicators:

1. **Equity-Adjusted Funding Ratio (EAFR)** – Solvency measured again differential contribution patterns and longevity.
2. **Climate-Adjusted Liability Duration (CALD)** – The shift in liability horizons under warming scenarios.
3. **Intergenerational Proportionality Index (IPI)** – Fairness tracked as benefit-to-contribution ratios across cohorts.

Boards that can disclose these metrics will earn lower funding costs and higher trust; those that cannot will face widening spreads and regulatory intervention.

Conclusion: Fiduciary Realism

Pension systems were built for a demographic pyramid that no longer exists and a climate that no longer holds steady. The actuarial math still works, but governance does not.

Empathy won't close funding gaps. Information will. Systems that treat inequality, health variance, and climate exposure as data—not politics—remain solvent.

Governance is the last functioning translation device between uncertainty and stability. Pension boards, whether they recognize it or not, are making the time value of trust itself.

This series explores ideas from “[Shaping the Decade: Governance, Sustainability, and AI 2026–2036](#),” a guide for boards at the crossroads of governance, technology, and stakeholder capitalism. [Available Here](#).

Tanya Matanda is a governance strategist bridging institutional oversight, governance, and fiduciary resilience. Her work supports boards, LPs, and regulators in designing governance systems fit for the AI era.

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